Railway Transport Database System

(RTDS)

Dhirubhai Ambani University

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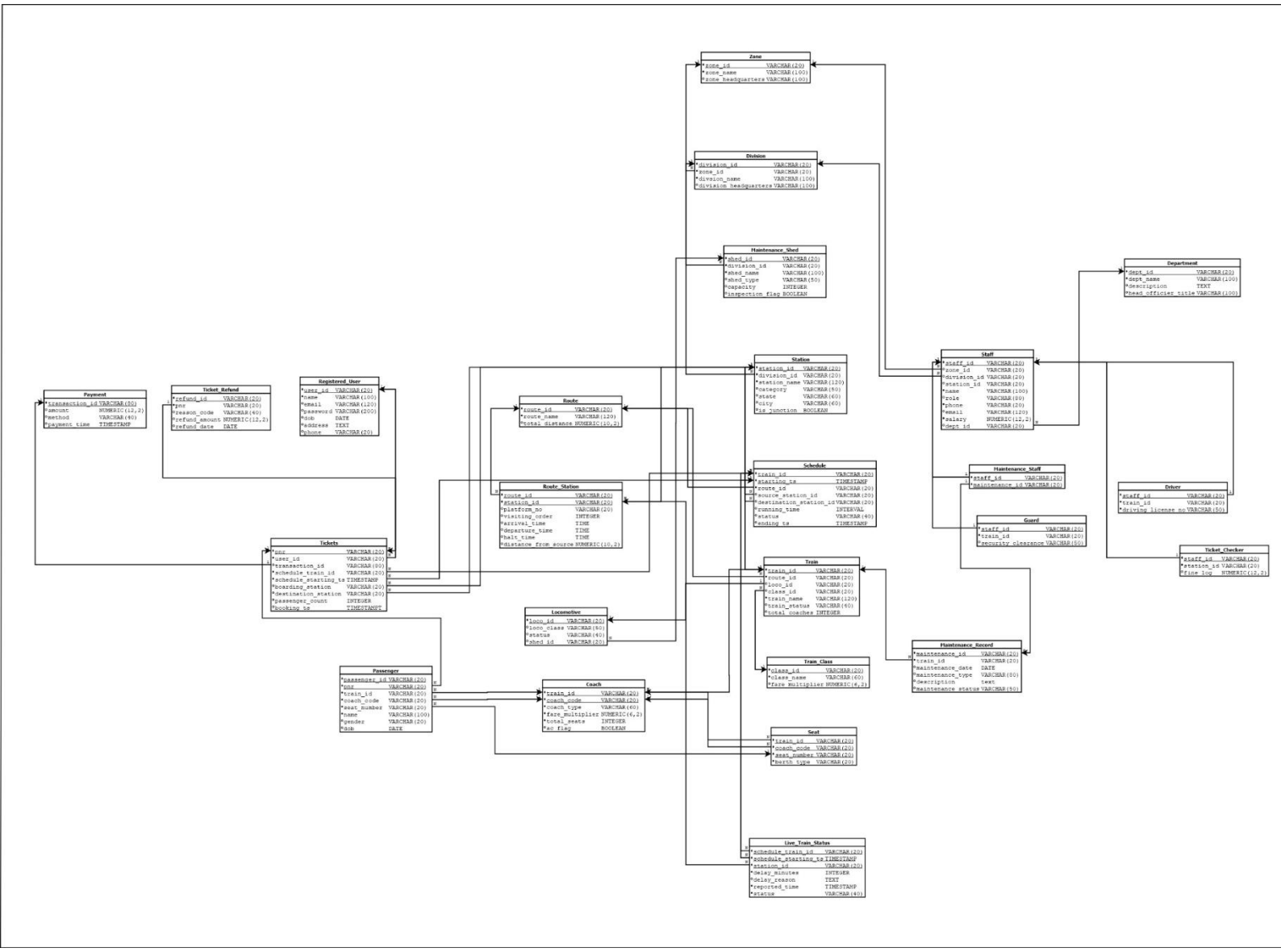
- **Relational Schema Diagram**: -



- **Minimal FD Set**: -

1. ZONE

zone_id → zone_name

zone_id → zone_headquarters

2. DIVISION

division_id → zone_id

division_id → division_name

division_id → division_headquarters

3. DEPARTMENT

dept_id → dept_name

dept_id → description

dept_id → head_officer_title

4. STATION

station_id → division_id

station_id → station_name

station_id → category

station_id → state

station_id → city

station_id → is_junction

5. ROUTE

route_id → route_name

route_id → total_distance

6. TRAIN_CLASS

class_id → class_name

class_id → fare_multiplier

7. REGISTERED_USER

user_id → name

user_id → email

user_id → password

user_id → dob

user_id → address
user_id → phone
email → user_id
phone → user_id

email → user_id and phone → user_id are kept because email and phone are UNIQUE, removing them would lose the bidirectional determination between the candidate keys.

8. MAINTENANCE_SHED
shed_id → division_id
shed_id → shed_name
shed_id → shed_type
shed_id → capacity
shed_id → inspection_flag

9. LOCOMOTIVE
loco_id → loco_class
loco_id → status
loco_id → shed_id

10. TRAIN
train_id → route_id
train_id → loco_id
train_id → train_name
train_id → train_status
train_id → total_coaches
train_id → class_id
train_id → base_fare

11. MAINTENANCE_RECORD
maintenance_id → train_id
maintenance_id → maintenance_date
maintenance_id → maintenance_type
maintenance_id → description
maintenance_id → maintenance_status

12. STAFF
staff_id → zone_id
staff_id → division_id
staff_id → station_id
staff_id → name
staff_id → role
staff_id → phone
staff_id → email
staff_id → salary
staff_id → dept_id
email → staff_id
phone → staff_id

email → staff_id and phone → staff_id are kept because email and phone are UNIQUE.

13. MAINTENANCE_STAFF
No non-trivial FDs — relation has only key attributes (staff_id, maintenance_id).

14. TICKET_CHECKER
staff_id → station_id
staff_id → fine_log

15. DRIVER
staff_id → train_id
staff_id → driving_license_no
driving_license_no → staff_id

driving_license_no → staff_id is non-redundant because driving_license_no is a UNIQUE.

16. GUARD
staff_id → train_id
staff_id → security_clearance

17. ROUTE_STATION
(route_id, station_id) → platform_no
(route_id, station_id) → visiting_order
(route_id, station_id) → arrival_time
(route_id, station_id) → departure_time
(route_id, station_id) → halt_time
(route_id, station_id) → distance_from_source

18. SCHEDULE
(train_id, starting_ts) → running_time
(train_id, starting_ts) → status
(train_id, starting_ts) → ending_ts
(train_id, starting_ts) → route_id
(train_id, starting_ts) → source_station_id
(train_id, starting_ts) → destination_station_id

19. PAYMENT
transaction_id → amount
transaction_id → method
transaction_id → payment_time

20. TICKETS
pnr → passenger_count
pnr → booking_ts
pnr → user_id
pnr → transaction_id
pnr → boarding_station
pnr → destination_station
pnr → schedule_train_id
pnr → schedule_starting_ts

21. COACH
(train_id, coach_code) → coach_type
(train_id, coach_code) → fare_multiplier
(train_id, coach_code) → total_seats
(train_id, coach_code) → ac_flag

22. SEAT
(train_id, coach_code, seat_number) → berth_type

23. PASSENGER
(passenger_id, pnr) → train_id
(passenger_id, pnr) → coach_code
(passenger_id, pnr) → seat_number
(passenger_id, pnr) → name
(passenger_id, pnr) → gender
(passenger_id, pnr) → dob

24. TICKET_REFUND
refund_id → pnr
refund_id → reason_code
refund_id → refund_amount
refund_id → refund_date

25. LIVE_TRAIN_STATUS
(schedule_train_id, schedule_starting_ts, station_id) → delay_minutes
(schedule_train_id, schedule_starting_ts, station_id) → delay_reason
(schedule_train_id, schedule_starting_ts, station_id) → reported_time
(schedule_train_id, schedule_starting_ts, station_id) → status

- Proof that relations are in BCNF: -

1. ZONE

- **Attributes:** zone_id, zone_name, zone_headquarters
- **Candidate Keys:** {zone_id}
- **All FDs:**
  - zone_id → zone_name
  - zone_id → zone_headquarters
- **BCNF Check:** For every FD, the determinant is zone_id. Is zone_id a superkey? Yes — it is the sole candidate key and determines all attributes.
- **Result:** zone is in BCNF.

2. DIVISION

- **Attributes:** division_id, zone_id, division_name, division_headquarters
- **Candidate Keys:** {division_id}
- **All FDs:**
  - division_id → zone_id
  - division_id → division_name
  - division_id → division_headquarters
- **BCNF Check:** All FDs have division_id as determinant, which is the sole candidate key.
- **Result:** division is in BCNF.

3. DEPARTMENT

- **Attributes:** dept_id, dept_name, description, head_officer_title

- **Candidate Keys:** {dept_id}
- **All FDs:**
  - dept_id → dept_name
  - dept_id → description
  - dept_id → head_officer_title
- **BCNF Check:** All FDs have dept_id as determinant, which is the sole candidate key.
- **Result:** department is in BCNF.

4. STATION

- **Attributes:** station_id, division_id, station_name, category, state, city, is_junction
- **Candidate Keys:** {station_id}
- **All FDs:**
  - station_id → division_id
  - station_id → station_name
  - station_id → category
  - station_id → state
  - station_id → city
  - station_id → is_junction
- **BCNF Check:** All FDs have station_id as determinant, which is the sole candidate key. No non-key attribute determines another.
- **Result:** station is in BCNF.

5. ROUTE

- **Attributes:** route_id, route_name, total_distance

- **Candidate Keys:** {route_id}
- **All FDs:**
    - route_id → route_name
    - route_id → total_distance
- **BCNF Check:** All FDs have route_id as determinant, which is the sole candidate key.
- **Result:** route is in BCNF.

6. TRAIN_CLASS

- **Attributes:** class_id, class_name, fare_multiplier
- **Candidate Keys:** {class_id}
- **All FDs:**
    - class_id → class_name
    - class_id → fare_multiplier
- **BCNF Check:** All FDs have class_id as determinant, which is the sole candidate key.
- **Result:** train_class is in BCNF.

7. REGISTERED_USER

- **Attributes:** user_id, name, email, password, dob, address, phone
- **Candidate Keys:** {user_id}, {email}, {phone}
- **All FDs:**
    - user_id → name, email, password, dob, address, phone
    - email → user_id
    - phone → user_id

- **BCNF Check:** Every non-trivial FD has a determinant that is a candidate key:
    - user_id → {all other attributes} — user_id is a candidate key
    - email → user_id — email is a candidate key
    - phone → user_id — phone is a candidate key
    - By transitivity: email → {name, password, dob, address, phone} and phone → {name, email, password, dob, address} — both determinants are candidate keys.
    - No partial dependencies (none of the candidate keys is composite).
    - No transitive dependency — no non-key attribute determines another non-key attribute.
- **Result:** registered_user is in BCNF.

8. MAINTENANCE_SHED

- **Attributes:** shed_id, division_id, shed_name, shed_type, capacity, inspection_flag
- **Candidate Keys:** {shed_id}
- **All FDs:**
    - shed_id → division_id
    - shed_id → shed_name
    - shed_id → shed_type
    - shed_id → capacity
    - shed_id → inspection_flag
- **BCNF Check:** All FDs have shed_id as determinant, which is the sole candidate key.

- **Result:** maintenance_shed is in BCNF.

9. LOCOMOTIVE

- **Attributes:** loco_id, loco_class, status, shed_id
- **Candidate Keys:** {loco_id}
- **All FDs:**
  - loco_id → loco_class
  - loco_id → status
  - loco_id → shed_id
- **BCNF Check:** All FDs have loco_id as determinant, which is the sole candidate key.
- **Result:** locomotive is in BCNF.

10. TRAIN

- **Attributes:** train_id, route_id, loco_id, train_name, train_status, total_coaches, class_id, base_fare
- **Candidate Keys:** {train_id}
- **All FDs:**
  - train_id → route_id
  - train_id → loco_id
  - train_id → train_name
  - train_id → train_status
  - train_id → total_coaches
  - train_id → class_id
  - train_id → base_fare
- **BCNF Check:** All FDs have train_id as determinant, which is the sole candidate key. Foreign key attributes (route_id, loco_id,

class_id) are facts about the train itself, not determinants of other attributes.

- **Result:** train is in BCNF.

11. MAINTENANCE_RECORD

- **Attributes:** maintenance_id, train_id, maintenance_date, maintenance_type, description, maintenance_status
- **Candidate Keys:** {maintenance_id}
- **All FDs:**
    - maintenance_id → train_id
    - maintenance_id → maintenance_date
    - maintenance_id → maintenance_type
    - maintenance_id → description
    - maintenance_id → maintenance_status
- **BCNF Check:** All FDs have maintenance_id as determinant, which is the sole candidate key.
- **Result:** maintenance_record is in BCNF.

12. STAFF

- **Attributes:** staff_id, zone_id, division_id, station_id, name, role, phone, email, salary, dept_id
- **Candidate Keys:** {staff_id}, {email}, {phone}
- **All FDs:**
    - staff_id → zone_id, division_id, station_id, name, role, phone, email, salary, dept_id
    - email → staff_id
    - phone → staff_id

- **BCNF Check:** Every non-trivial FD has a determinant that is a candidate key:
    - staff_id → {all other attributes} — staff_id is a candidate key
    - email → staff_id — email is a candidate key
    - phone → staff_id — phone is a candidate key
    - By transitivity: email and phone determine all other attributes through staff_id.
    - No partial dependencies (none of the candidate keys is composite).
    - No transitive dependency among non-key attributes.
- **Result:** staff is in BCNF.

13. MAINTENANCE_STAFF

- **Attributes:** staff_id, maintenance_id
- **Candidate Keys:** {staff_id, maintenance_id}
- **All FDs:** No non-trivial FDs (only key attributes)
- **BCNF Check:** A relation with only key attributes trivially satisfies BCNF — there are no non-trivial FDs to violate it.
- **Result:** maintenance_staff is in BCNF.

14. TICKET_CHECKER

- **Attributes:** staff_id, station_id, fine_log
- **Candidate Keys:** {staff_id}
- **All FDs:**
    - staff_id → station_id
    - staff_id → fine_log

- **BCNF Check:** All FDs have staff_id as determinant, which is the sole candidate key.
- **Result:** ticket_checker is in BCNF.

15. DRIVER

- **Attributes:** staff_id, train_id, driving_license_no
- **Candidate Keys:** {staff_id}, {driving_license_no}
- **All FDs:**
  - staff_id → train_id
  - staff_id → driving_license_no
  - driving_license_no → staff_id
- **BCNF Check:** Every non-trivial FD has a determinant that is a candidate key:
  - staff_id → train_id, driving_license_no — staff_id is a candidate key
  - driving_license_no → staff_id — driving_license_no is a candidate key
  - By transitivity: driving_license_no → train_id — determinant is a candidate key.
  - No partial dependencies (neither candidate key is composite).
  - No transitive dependency.
- **Result:** driver is in BCNF.

16. GUARD

- **Attributes:** staff_id, train_id, security_clearance
- **Candidate Keys:** {staff_id}

- **All FDs:**
  - staff_id → train_id
  - staff_id → security_clearance
- **BCNF Check:** All FDs have staff_id as determinant, which is the sole candidate key.
- **Result:** guard is in BCNF.

17. ROUTE_STATION

- **Attributes:** route_id, station_id, platform_no, visiting_order, arrival_time, departure_time, halt_time, distance_from_source
- **Candidate Keys:** {route_id, station_id}
- **All FDs:**
  - (route_id, station_id) → platform_no
  - (route_id, station_id) → visiting_order
  - (route_id, station_id) → arrival_time
  - (route_id, station_id) → departure_time
  - (route_id, station_id) → halt_time
  - (route_id, station_id) → distance_from_source
- **BCNF Check:** All FDs have the composite PK {route_id, station_id} as determinant, which is the sole candidate key (and thus a superkey). Partial dependency check:
  - route_id alone → platform_no No — same route has different platforms at different stations.
  - station_id alone → platform_no No — same station has different platforms for different routes.
  - No partial or transitive dependencies exist.

- **Result:** route_station is in BCNF.

18. SCHEDULE

- **Attributes:** train_id, starting_ts, running_time, status, ending_ts, route_id, source_station_id, destination_station_id
- **Candidate Keys:** {train_id, starting_ts}
- **All FDs:**
  - (train_id, starting_ts) → running_time
  - (train_id, starting_ts) → status
  - (train_id, starting_ts) → ending_ts
  - (train_id, starting_ts) → route_id
  - (train_id, starting_ts) → source_station_id
  - (train_id, starting_ts) → destination_station_id
- **BCNF Check:** All FDs have the composite PK {train_id, starting_ts} as determinant, which is the sole candidate key. Partial dependency check:
  - train_id alone → route_id? No — same train can run on different routes at different times.
  - starting_ts alone → anything? No — timestamp alone doesn't identify which train.
  - No partial or transitive dependencies.
- **Result:** schedule is in BCNF.

19. PAYMENT

- **Attributes:** transaction_id, amount, method, payment_time
- **Candidate Keys:** {transaction_id}
- **All FDs:**

    - transaction_id → amount
    - transaction_id → method
    - transaction_id → payment_time
- **BCNF Check:** All FDs have transaction_id as determinant, which is the sole candidate key.
- **Result:** payment is in BCNF.

20. TICKETS

- **Attributes:** pnr, passenger_count, booking_ts, user_id, transaction_id, boarding_station, destination_station, schedule_train_id, schedule_starting_ts
- **Candidate Keys:** {pnr}
- **All FDs:**
    - pnr → passenger_count
    - pnr → booking_ts
    - pnr → user_id
    - pnr → transaction_id
    - pnr → boarding_station
    - pnr → destination_station
    - pnr → schedule_train_id
    - pnr → schedule_starting_ts
- **BCNF Check:** All FDs have pnr as determinant, which is the sole candidate key. Foreign key attributes are facts about this ticket, not determinants.
- **Result:** tickets is in BCNF.

21. PASSENGER

- **Attributes:** passenger_id, pnr, train_id, coach_code, seat_number, name, gender, dob
- **Candidate Keys:** {passenger_id, pnr}
- **All FDs:**
  - (passenger_id, pnr) → train_id
  - (passenger_id, pnr) → coach_code
  - (passenger_id, pnr) → seat_number
  - (passenger_id, pnr) → name
  - (passenger_id, pnr) → gender
  - (passenger_id, pnr) → dob
- **BCNF Check:** All FDs have the composite PK {passenger_id, pnr} as determinant, which is the sole candidate key. Partial dependency check:
  - passenger_id alone → name? No — same person can have different names on different bookings (or be listed multiple times).
  - pnr alone → train_id? No — one PNR can have multiple passengers.
  - No partial or transitive dependencies.
- **Result:** passenger is in BCNF.

22. COACH

- **Attributes:** coach_code, train_id, coach_type, fare_multiplier, total_seats, ac_flag
- **Candidate Keys:** {coach_code, train_id}
- **All FDs:**
  - (coach_code, train_id) → coach_type

    - (coach_code, train_id) → fare_multiplier
    - (coach_code, train_id) → total_seats
    - (coach_code, train_id) → ac_flag
- **BCNF Check:** All FDs have the composite PK {coach_code, train_id} as determinant, which is the sole candidate key. Partial dependency check:
    - coach_code alone → coach_type? No — same coach code can have different types on different trains.
    - train_id alone → coach_type? No — same train has multiple coaches of different types.
    - No partial or transitive dependencies.
- **Result:** coach is in BCNF.

23. SEAT

- **Attributes:** train_id, coach_code, seat_number, berth_type
- **Candidate Keys:** {train_id, coach_code, seat_number}
- **All FDs:**
    - (train_id, coach_code, seat_number) → berth_type
- **BCNF Check:** The only non-trivial FD has the composite PK {train_id, coach_code, seat_number} as determinant, which is the sole candidate key. Partial dependency check:
    - Can any proper subset of the key determine berth_type? No — you need all three to uniquely identify a seat and its berth type.
    - No transitive dependencies (only one non-key attribute).
- **Result:** seat is in BCNF.

24. TICKET_REFUND

- **Attributes:** refund_id, pnr, reason_code, refund_amount, refund_date
- **Candidate Keys:** {refund_id}
- **All FDs:**
  - refund_id → pnr
  - refund_id → reason_code
  - refund_id → refund_amount
  - refund_id → refund_date
- **BCNF Check:** All FDs have refund_id as determinant, which is the sole candidate key.
- **Result:** ticket_refund is in BCNF.

25. LIVE_TRAIN_STATUS

- **Attributes:** schedule_train_id, schedule_starting_ts, station_id, delay_minutes, delay_reason, reported_time, status
- **Candidate Keys:** {schedule_train_id, schedule_starting_ts, station_id}
- **All FDs:**
  - (schedule_train_id, schedule_starting_ts, station_id) → delay_minutes
  - (schedule_train_id, schedule_starting_ts, station_id) → delay_reason
  - (schedule_train_id, schedule_starting_ts, station_id) → reported_time
  - (schedule_train_id, schedule_starting_ts, station_id) → status

- **BCNF Check:** All FDs have the composite PK {schedule_train_id, schedule_starting_ts, station_id} as determinant, which is the sole candidate key. Partial dependency check:
    - (schedule_train_id, schedule_starting_ts) → delay_minutes? No — delay is specific to each station on the journey.
    - station_id → delay_minutes? No — different trains have different delays at the same station.
    - No partial or transitive dependencies.

Result: live_train_status is in BCNF.

- DDL Scripts: -

```
CREATE SCHEMA IF NOT EXISTS DB_Project;

SET SEARCH_PATH TO DB_Project;


CREATE TABLE zone (
   zone_id VARCHAR(20) PRIMARY KEY,
   zone_name VARCHAR(100) NOT NULL,
   zone_headquarters VARCHAR(100) NOT NULL
);

CREATE TABLE division (
```

```
    division_id VARCHAR(20) PRIMARY KEY,
    zone_id VARCHAR(20) NOT NULL REFERENCES zone(zone_id)
ON UPDATE CASCADE ON DELETE RESTRICT,
    division_name VARCHAR(100) NOT NULL,
    division_headquarters VARCHAR(100) NOT NULL
);

CREATE TABLE department (
    dept_id VARCHAR(20) PRIMARY KEY,
    dept_name VARCHAR(100) NOT NULL,
    description TEXT,
    head_officer_title VARCHAR(100) NOT NULL
);




CREATE TABLE station (
    station_id VARCHAR(20) PRIMARY KEY,
    division_id VARCHAR(20) NOT NULL REFERENCES
division(division_id) ON UPDATE CASCADE ON DELETE
RESTRICT,
    station_name VARCHAR(120) NOT NULL,
    category VARCHAR(50) NOT NULL,
    state VARCHAR(60) NOT NULL,
    city VARCHAR(60) NOT NULL,
    is_junction BOOLEAN DEFAULT FALSE
);

CREATE TABLE route (
    route_id VARCHAR(20) PRIMARY KEY,
```

```
    route_name VARCHAR(120) NOT NULL,
    total_distance NUMERIC(10,2) NOT NULL CHECK
(total_distance >= 0)
);

CREATE TABLE train_class (
    class_id VARCHAR(20) PRIMARY KEY,
    class_name VARCHAR(60) NOT NULL,
    fare_multiplier NUMERIC(6,2) NOT NULL DEFAULT 1.0 CHECK
(fare_multiplier > 0)
);

CREATE TABLE registered_user (
    user_id VARCHAR(20) PRIMARY KEY,
    name VARCHAR(100) NOT NULL,
    email VARCHAR(120) UNIQUE NOT NULL,
    password VARCHAR(200) NOT NULL,
    dob DATE NOT NULL,
    address TEXT NOT NULL,
    phone VARCHAR(20) UNIQUE NOT NULL
);

CREATE TABLE maintenance_shed (
    shed_id VARCHAR(20) PRIMARY KEY,
    division_id VARCHAR(20) NOT NULL REFERENCES
division(division_id) ON UPDATE CASCADE ON DELETE
RESTRICT,
    shed_name VARCHAR(100) NOT NULL,
    shed_type VARCHAR(50) NOT NULL,
    capacity INTEGER NOT NULL CHECK (capacity >= 0),
    inspection_flag BOOLEAN DEFAULT FALSE
);
```

```
CREATE TABLE locomotive (
   loco_id VARCHAR(20) PRIMARY KEY,
   loco_class VARCHAR(50) NOT NULL,
   status VARCHAR(40) NOT NULL DEFAULT 'Active',
   shed_id VARCHAR(20) REFERENCES
maintenance_shed(shed_id) ON UPDATE CASCADE ON DELETE
SET NULL
);

CREATE TABLE train (
   train_id VARCHAR(20) PRIMARY KEY,
   route_id VARCHAR(20) NOT NULL REFERENCES
route(route_id) ON UPDATE CASCADE ON DELETE RESTRICT,
   loco_id VARCHAR(20) REFERENCES locomotive(loco_id) ON
UPDATE CASCADE ON DELETE SET NULL,
   class_id VARCHAR(20) NOT NULL REFERENCES
train_class(class_id) ON UPDATE CASCADE ON DELETE SET
NULL,
   train_name VARCHAR(120) NOT NULL,
   train_type VARCHAR(60) NOT NULL,
   train_status VARCHAR(40) NOT NULL DEFAULT 'Active',
   total_coaches INTEGER NOT NULL DEFAULT 0 CHECK
(total_coaches >= 0),
   base_fare NUMERIC(10,2) NOT NULL DEFAULT 0.00 CHECK
(base_fare >= 0)
);


CREATE TABLE maintenance_record (
   maintenance_id VARCHAR(20) PRIMARY KEY,
   train_id VARCHAR(20) NOT NULL REFERENCES train(train_id)
ON UPDATE CASCADE ON DELETE CASCADE,
   maintenance_date DATE NOT NULL DEFAULT CURRENT_DATE,
```

```
    maintenance_type VARCHAR(80) NOT NULL,
    description TEXT,
    maintenance_status VARCHAR(50) NOT NULL
);

CREATE TABLE staff (
    staff_id VARCHAR(20) PRIMARY KEY,
    zone_id VARCHAR(20) NOT NULL REFERENCES zone(zone_id),
    division_id VARCHAR(20) NOT NULL REFERENCES
division(division_id),
    station_id VARCHAR(20) REFERENCES station(station_id),
    dept_id VARCHAR(20) NOT NULL REFERENCES
department(dept_id),
    name VARCHAR(100) NOT NULL,
    role VARCHAR(80) NOT NULL,
    phone VARCHAR(20) UNIQUE NOT NULL,
    email VARCHAR(120) UNIQUE NOT NULL,
    salary NUMERIC(12,2) NOT NULL CHECK (salary >= 0)
);

CREATE TABLE maintenance_staff (
    staff_id VARCHAR(20) NOT NULL REFERENCES staff(staff_id)
ON DELETE CASCADE,
    maintenance_id VARCHAR(20) NOT NULL REFERENCES
maintenance_record(maintenance_id) ON DELETE CASCADE,
    PRIMARY KEY (staff_id, maintenance_id)
);

CREATE TABLE ticket_checker (
    staff_id VARCHAR(20) PRIMARY KEY REFERENCES
staff(staff_id) ON DELETE CASCADE,
    station_id VARCHAR(20) NOT NULL REFERENCES
station(station_id),
```

```
    fine_log NUMERIC(12,2) DEFAULT 0.00 CHECK (fine_log >= 0)
);

CREATE TABLE driver (
    staff_id VARCHAR(20) PRIMARY KEY REFERENCES
staff(staff_id) ON DELETE CASCADE,
    train_id VARCHAR(20) NOT NULL REFERENCES train(train_id),
    driving_license_no VARCHAR(50) UNIQUE NOT NULL
);

CREATE TABLE guard (
    staff_id VARCHAR(20) PRIMARY KEY REFERENCES
staff(staff_id) ON DELETE CASCADE,
    train_id VARCHAR(20) NOT NULL REFERENCES train(train_id),
    security_clearance VARCHAR(50) NOT NULL
);



CREATE TABLE route_station (
    route_id VARCHAR(20) NOT NULL REFERENCES
route(route_id) ON DELETE CASCADE,
    station_id VARCHAR(20) NOT NULL REFERENCES
station(station_id),
    platform_no VARCHAR(20),
    visiting_order INTEGER NOT NULL,
    arrival_time TIME,
    departure_time TIME,
    halt_time TIME DEFAULT '00:00:00',
    distance_from_source NUMERIC(10,2) NOT NULL CHECK
(distance_from_source >= 0),
    PRIMARY KEY (route_id, station_id)
);
```

```
CREATE TABLE schedule (
   train_id VARCHAR(20) NOT NULL REFERENCES train(train_id)
ON DELETE CASCADE,
   starting_ts TIMESTAMP NOT NULL,
   route_id VARCHAR(20) NOT NULL REFERENCES
route(route_id),
   source_station_id VARCHAR(20) NOT NULL REFERENCES
station(station_id),
   destination_station_id VARCHAR(20) NOT NULL REFERENCES
station(station_id),
   running_time INTERVAL,
   status VARCHAR(40) NOT NULL DEFAULT 'Scheduled',
   ending_ts TIMESTAMP,
   PRIMARY KEY (train_id, starting_ts)
);



CREATE TABLE payment (
   transaction_id VARCHAR(80) PRIMARY KEY,
   amount NUMERIC(12,2) NOT NULL CHECK (amount >= 0),
   method VARCHAR(40) NOT NULL,
   payment_time TIMESTAMP NOT NULL DEFAULT
CURRENT_TIMESTAMP
);

CREATE TABLE tickets (
   pnr VARCHAR(20) PRIMARY KEY,
   user_id VARCHAR(20) NOT NULL REFERENCES
registered_user(user_id),
   transaction_id VARCHAR(80) NOT NULL REFERENCES
payment(transaction_id),
```

```
    boarding_station VARCHAR(20) NOT NULL REFERENCES
station(station_id),
    destination_station VARCHAR(20) NOT NULL REFERENCES
station(station_id),
    schedule_train_id VARCHAR(20) NOT NULL,
    schedule_starting_ts TIMESTAMP NOT NULL,
    passenger_count INTEGER NOT NULL CHECK
(passenger_count > 0),
    booking_ts TIMESTAMP NOT NULL DEFAULT
CURRENT_TIMESTAMP,
    CONSTRAINT fk_ticket_schedule FOREIGN KEY
(schedule_train_id, schedule_starting_ts)
        REFERENCES schedule(train_id, starting_ts)
);




CREATE TABLE coach (
    train_id VARCHAR(20) NOT NULL REFERENCES train(train_id)
ON DELETE CASCADE,
    coach_code VARCHAR(20) NOT NULL,
    coach_type VARCHAR(60) NOT NULL,
    fare_multiplier NUMERIC(6,2) NOT NULL DEFAULT 1.0 CHECK
(fare_multiplier > 0),
    total_seats INTEGER NOT NULL CHECK (total_seats >= 0),
    ac_flag BOOLEAN DEFAULT FALSE,
    PRIMARY KEY (train_id, coach_code)
);

CREATE TABLE seat (
    train_id VARCHAR(20) NOT NULL,
```

```
   coach_code VARCHAR(20) NOT NULL,
   seat_number VARCHAR(20) NOT NULL,
   berth_type VARCHAR(20) NOT NULL,
   PRIMARY KEY (train_id, coach_code, seat_number),
   CONSTRAINT fk_seat_coach FOREIGN KEY (train_id,
coach_code)
       REFERENCES coach(train_id, coach_code) ON DELETE
CASCADE
);

CREATE TABLE passenger (
   passenger_id VARCHAR(20) NOT NULL,
   pnr VARCHAR(20) NOT NULL REFERENCES tickets(pnr) ON
DELETE CASCADE,
   train_id VARCHAR(20) NOT NULL,
   coach_code VARCHAR(20) NOT NULL,
   seat_number VARCHAR(20) NOT NULL,
   name VARCHAR(100) NOT NULL,
   gender VARCHAR(20) NOT NULL,
   dob DATE NOT NULL,
   berth_pref VARCHAR(30),
   PRIMARY KEY (passenger_id, pnr),
   CONSTRAINT fk_passenger_seat FOREIGN KEY (train_id,
coach_code, seat_number)
       REFERENCES seat(train_id, coach_code, seat_number)
);


CREATE TABLE ticket_refund (
   refund_id VARCHAR(20) PRIMARY KEY,
   pnr VARCHAR(20) NOT NULL REFERENCES tickets(pnr) ON
DELETE CASCADE,
   reason_code VARCHAR(40) NOT NULL,
```

```
    refund_amount NUMERIC(12,2) NOT NULL CHECK
(refund_amount >= 0),
    refund_date DATE NOT NULL DEFAULT CURRENT_DATE
);

CREATE TABLE live_train_status (
    schedule_train_id VARCHAR(20) NOT NULL,
    schedule_starting_ts TIMESTAMP NOT NULL,
    station_id VARCHAR(20) NOT NULL REFERENCES
station(station_id),
    delay_minutes INTEGER NOT NULL DEFAULT 0 CHECK
(delay_minutes >= 0),
    delay_reason TEXT,
    reported_time TIMESTAMP NOT NULL DEFAULT
CURRENT_TIMESTAMP,
    status VARCHAR(40) NOT NULL,
    PRIMARY KEY (schedule_train_id, schedule_starting_ts,
station_id),
    CONSTRAINT fk_lts_schedule FOREIGN KEY
(schedule_train_id, schedule_starting_ts)
        REFERENCES schedule(train_id, starting_ts) ON DELETE
CASCADE
);
```

To look at DIA properly refer to this Github Link:-
https://github.com/RonakVataliya/Railway_Transport_Database_System_RTDS